# Tanay Chitlur

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## Education

**Carnegie Mellon University** — Pittsburgh, PA
*Bachelor of Science in Computer Science — GPA: 4.0/4.0* — *Expected May 2029*
- Relevant Coursework: Discrete Math, Principles of Imperative Computation, Linear Algebra, Calculus 1–3.

## Research and Publications

**A Morphologically-Aware Transformer Network for High-Fidelity V1–V5 ECG Lead Generation** — October 2025
- IEEE Sensors 2025 (24th annual): First-authored and presented a peer-reviewed paper; full text in IEEE Xplore.
- Designed and trained a multi-stage transformer in Python with custom attention mechanisms, a mixture-of-experts decoder, and an eight-term composite loss function; built the full pipeline end-to-end, from data preprocessing and augmentation of 22K+ signal samples to model training, evaluation, and validation.

**Transformer-Based Framework with Dynamic Multiscale Attention Mechanisms for 12-Lead ECG Synthesis** — July 2025
- IEEE EMBC 2025 (47th annual): First-authored and presented a peer-reviewed paper at the largest biomedical engineering conference; full text in IEEE Xplore and PubMed.
- Engineered a novel transformer in PyTorch with saliency-driven adaptive attention windows, multi-scale feature fusion, and a custom SMAPE loss for time-series generation, enabling full 12-lead ECGs from a consumer wearable; an earlier iteration earned an International Science and Engineering Fair (ISEF) Grand Award and was presented at HRX 2024.

**IntelliCane: A Deep Learning Cane for Active Obstacle Avoidance & Navigation for the Visually Impaired** — May 2025
- Built a servo-guided smart cane end-to-end, from custom hardware to an on-device vision and navigation stack that actively steers visually impaired users; won the ISEF 3rd Place Grand Award and AAAI Special Award at Regeneron ISEF.
- Engineered a real-time embedded system integrating computer vision (depth estimation, YOLO object detection), A* pathfinding, and LLM-based decision-making; compressed models via ONNX export and INT8 quantization to cut inference ∼17x, delivering sub-second end-to-end latency on a Raspberry Pi at a $207 total build cost.

**Conference Presentations:** IEEE Sensors 2025, IEEE EMBC 2025, Regeneron ISEF 2024 & 2025, HRX 2024, Heart Rhythm Society Annual Meeting (HRS'24), AI Med 2023, Baylor Scott & White Tech4Better Symposium 2025

## Professional Experience

**Machine Learning Engineering Intern** — June 2024 – July 2026
*Hello Alfred.ai* — *Austin, TX*
- Built an ML-driven medical history bot for an AI health companion using Python, LangChain, GPT-4, and Pinecone, triaging patient history to flag when clinical care is needed; deployed to 10K+ users, resolving 85% of intake autonomously.
- Cut the time to process and distill a patient's medical history from 14 to 4 minutes and raised triage accuracy to 93% via retrieval-augmented prompting; presented results at the 2025 Baylor Scott & White Tech4Better Symposium.

**Team Captain & Software Lead** — November 2021 – March 2025
*FIRST Tech Challenge — Team 18108, High Voltage* — *Portland, OR*
- Led a 12-student team; wrote autonomous software with OpenCV vision and RoadRunner motion planning (PID control), raising autonomous scoring 32%; won the Control Award at the Oregon State Championship (1st of 40+ teams).

**Founder** — July 2019 – Present
*TnT Treats® Baking LLC* — *Portland, OR*
- Founded and scaled an artisanal dessert business to $25K+ in lifetime sales across 1,500+ custom orders; own pricing, marketing, and fulfillment end-to-end, with 70% of revenue from repeat customers.

## Projects

**Slipstream: Rare-Disease Research Navigator** — Hack-Nation Global AI Hackathon, 2nd Place of 3,000+ — October 2026
- Built a 2,700-node disease graph with Louvain clustering and created an LLM pipeline processing 1,000+ studies & trials.

## Awards

**Coca-Cola Scholar**, 2026 — $20,000 merit-based national scholarship; 1 of 150 selected from 110,000+ applicants
**HackCMU**, 2nd Place — Food Track, 2026
**ISEF Grand Award**, 3rd Place & AAAI Special Award — Robotics and Intelligent Machines, Regeneron ISEF, 2025
**ISEF Grand Award**, 2nd Place — Robotics and Intelligent Machines, Regeneron ISEF, 2024

## Technical Skills

**Languages**: Python, C++, Java, Swift, JavaScript, CUDA, HTML
**ML & AI**: PyTorch, TensorFlow, Keras, Scikit-Learn, Hugging Face, XGBoost, ONNX, OpenCV, LangChain, Pinecone
**Data & Scientific**: Pandas, NumPy, SciPy, Matplotlib, Seaborn, h5py
**Tools & Platforms**: Git, Linux, AWS, React, Raspberry Pi, Fusion 360, Cursor, Claude Code, OpenAI Codex